

MODULATION METHODS EMPLOYED IN DIGITAL COMMUNICATION: An Analysis

Adeleke, Oluseye A. and Abolade, Robert O.

Abstract

The concept of modulation is a key factor in communication, because without an appropriate modulation method or scheme, getting an expected throughput in such a communication effort would be impossible to achieve. Communication engineers and researchers have not relented in trying to find the best modulation methods aimed at achieving an expected throughput, bandwidth/power efficiency, low error performance, etc in digital communication networks. Digital modulation is preferred to the analogue modulation due to their error-free capability. Moreover, this choice of digital modulation is dependent on the type of communication network that is to be established. However, trade-offs must be made between the bandwidth efficiency, power efficiency and the cost of implementation of such a network.

This paper focuses on some methods usually employed in the process of modulation in digital communication networks. Specifically, binary phase shift keying, quadrature phase shift keying, eight phase shift keying and sixteen phase shift keying are investigated, using the matrix laboratory (MATLAB) software as the simulation tool in which the bit error rate (BER), symbol error rate (SER), spectral efficiency and power efficiency are comparatively analysed. Findings show that lower end phase shift keying schemes can be used for purposes that involve low error performance and minimum power but low bandwidth efficiency while the higher schemes suit purposes that require higher bandwidth efficiency but are low power efficient.

Keywords: Modulation, Power efficiency, Noise Immunity, Phase Shift Keying, Spectral Efficiency, Bandwidth Efficiency

1. Introduction

The transition from analog to digital modulation provides more information capacity, compatibility with digital data services, higher data security, better quality communications, and quicker system availability [6]. However, developers of communication systems face such constraints as available bandwidth, permissible power and inherent noise level of the system

The RF spectrum must be shared, yet every day there are more users for that spectrum as demand for communications services increases. Digital modulation schemes have greater capacity to convey large amounts of information than analog modulation schemes [1], [4].

Digital modulation has inherent benefits over analogue modulation because its distinct transmission states can more easily be detected at a receiver in the presence of noise than an analogue signal, which can assume an infinite number of values. On the other hand, when a digitally transmitted signal originates as an analog waveform, a trade-off occurs since some information is always lost in the quantization process necessary to convert the analogue signal to a digital one. [4], [5]

The various modulation schemes offered different solutions in terms of cost-effectiveness and quality of received signal, bandwidth efficiency and power efficiency but until recently were still largely analog. Frequency modulation and phase modulation presented certain immunity to noise, whereas amplitude modulation was simpler to demodulate. However, more recently with the advent of

low-cost microcontrollers and the introduction of domestic mobile telephones and satellite communications, digital modulation has gained popularity. With digital modulation techniques come all the advantages that traditional microprocessor circuits have over their analog counterparts. Any shortfalls in the communication link can be eradicated using software. Information can now be encrypted, error correction can ensure more confidence in received data, and the use of digital signal processing (DSP) techniques can reduce the limited bandwidth allocated to each service [2].

Digital modulation schemes are classified based on the aforementioned ways of modulation the first of the schemes is the amplitude modulation which consists of the OOK [On and Off Keying] which is the simplest form of amplitude modulation which is also known as 2-ASK i.e. Amplitude Shift Keying that varies between two states. Higher-order ASK modulation scheme is QAM [Quadrature Amplitude Modulation] which has various sub-schemes like 8QAM, 16QAM, 32QAM, 64QAM etc. However amplitude modulation is very susceptible to noise interference.

As frequency and phase modulation techniques offer more immunity to noise, they are the preferred schemes for the majority of services in use today. Frequency modulation offers FSK [Frequency Shift Keying] that is divided into various schemes like: BPSK, QFSK, 8-FSK, 16-FSK etc. while the third digital modulating technique uses phase modulation and it is called 'Phase Shift Keying' which offers various schemes like: BPSK, DPSK, QPSK, 8-PSK, 16-PSK etc.

2. Phase Shift keying (PSK)

This is a digital modulation method that conveys data by changing the phase of a reference signal (the carrier wave). PSK uses a finite number of phases; each assigned a unique pattern of binary bits. Usually, each phase encodes an equal number of bits. Each pattern of bits forms the symbol that is represented by the particular phase. The demodulator, which is designed specifically for the symbol-set used by the modulator, determines the phase of the received signal and maps it back to the symbol it represents, thus recovering the original data. This requires the receiver to be able to compare the phase of the received signal with a reference signal. [7]

In the case of PSK, the phase is changed to represent the data signal. There are two fundamental ways of utilizing the phase of a signal in this way:

1. By viewing the phase itself as conveying the information, in which case the demodulator must have a reference signal to compare the received signal's phase against; or
2. By viewing the change in the phase as conveying information differential schemes, which do not need a reference carrier [3].

In the PSK schemes, the phase of the carrier takes on one of M possible values or symbols. The (stream of) input binary data is first divided into b -bit blocks. Each block is transmitted as one of M possible values or symbols. Each symbol is a carrier frequency sinusoid having one of M possible phase values spaced $2\pi/M$ apart. Then, for the n th M-PSK say,

The M-PSK phase of (equation 2.1) will exist in all intervals:

$$\theta_n \in (0, \frac{2\pi}{m}, \frac{4\pi}{m}, \dots, 2(m-1)\pi/m) \quad (1)$$

Therefore, the modulation angle is

$$\theta_i = \frac{2\pi i}{m} \quad (2)$$

where m stands for the order of the modulation; if $m = 2$ the scheme will be BPSK i.e. Binary phase shift keying, if $m = 4$, it will be QPSK i.e. Quadrature phase shift keying, if $m = 8$ it is 8-PSK and so on.

To ensure that each transmitted binary digit contains an integral number of cycles, the unmodulated carrier frequency

$$f_c = k_c/T_s; \quad (3)$$

where k_c is a fixed integer.

3. Methodology

This research work investigates performance and comparison between different PSK modulation schemes in digital communication systems in the presence of thermal noise which is modeled as Additive White Gaussian Noise (AWGN). The research work aims at the choice of a modulation scheme for a particular purpose and which is achieved by the comparison of major factors that are being considered in choosing a modulation scheme, which are power efficiency, immunity to noise, bandwidth efficiency and spectral efficiency.

The M-ary modulation schemes that were considered are Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK), Eight-Phase Shift Keying (8-PSK) and Sixteen-Phase Shift Keying (16-PSK). For each of these modulation signaling schemes, the Bit Error Rate performance, coding of data for modulation and transmitting power required are considered. The analysis was carried out by computer simulation using MATLAB 7.4.

3.1 Probability of Bit Error and Bit Error rate

In digital communications it is desirable to minimize the average probability of bit error at the receiver subject to the constraints on received power and channel bandwidth. The terms probability of bit error (P_e) and bit error rate (BER) are used interchangeably in the literature although they differ slightly in meaning in practice. P_e is a mathematical representation for the bit error rate for a given system. BER is an empirical record of a system's actual bit error performance. In essence, system performance can be quantified by first measuring the bit error rate (BER) and then comparing the BER with the expected probability of bit error.

Probability of bit error is a function of the carrier-to-noise power ratio (C/N). This noise is often taken as thermal noise, which can be expressed as:

$$N = kTB \quad (\text{watts}) \quad (4)$$

The carrier power is a function of bit energy and bit duration:

$$C = \frac{E_b}{T_b} \quad (\text{Watts}) \quad (5)$$

where k = Boltzmann's constant = 1.38×10^{-23} W/(Hz-K)

B = noise bandwidth, Hz.

T = system noise temperature, K (note that 0°Celsius or 273K).

E_b = energy of a single bit, J/bit

Noise power density N_0 is the thermal power N normalized to a 1-Hz bandwidth:

$$N_0 = N/B \quad (6)$$

Substituting equation (7) and equation (8):

$$\begin{aligned} (E_b/N_0) &= CT_b/(N/B) \\ &= \left(\frac{C}{N}\right) \left(\frac{B}{r_b}\right) \\ \text{where } r_b &= \frac{1}{T_b} \end{aligned} \quad (7)$$

If the bandwidth equals the bit rate, the energy per bit-to-noise power density ratio will equal the carrier-to-noise power ratio, that is,

$$\frac{E_b}{N_0} = \frac{C}{N} \quad (8)$$

The general expression for the bit error probability for an M-PSK system is written as

$$P_e = \frac{1}{b} \operatorname{erfc} \left(\sqrt{b} \left(\frac{\sqrt{E_b}}{N_0} \right) \sin [\pi/2^b] \right) \quad (9)$$

where erfc denotes the complimentary error function. The larger the erf 's argument, the smaller the probability of error. Error function, erf , is, by definition,

$$\operatorname{erf}(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-t^2} dt \quad (10)$$

The complimentary error function is

$$\operatorname{erfc}(u) = 1 - \operatorname{erf}(u) \quad (11)$$

The integrals for these functions require numerical evaluation. For a large argument u , we can use a series expansion of the Laplace–Gauss integral of an asymptotic expansion:

$$\int_0^u e^{-t^2} dt = \frac{\sqrt{\pi}}{2} \left[1 - \frac{1}{2u^2} + \frac{1.3}{(2u^2)^2} - \frac{1.35}{(2u^2)^3} \right] \quad (12a)$$

Substituting equation (13) in equation (14a)

$$\int_0^u e^{-t^2} dt = \frac{e^{-u^2}}{u\sqrt{\pi}} \left[1 - \frac{1}{2u^2} + \frac{1.3}{(2u^2)^2} - \frac{1.3.5}{(2u^2)^3} + \dots \right] \quad (12b)$$

A frequently used function is the Q function, which is related to the *erfc* function by

$$Q(u) \cong \frac{1}{\sqrt{2\pi}} \int_u^\infty e^{-x^2/2} dx \quad (13)$$

Comparing the equation (14) and equation (15)

$$Q(u) = \frac{1}{2} \operatorname{erfc} \left(\frac{u}{\sqrt{2}} \right) \quad (14)$$

3.2 Channel Capacity

The rate at which the bits of digital information are encoded on each transmitted symbol determine the spectral efficiency of a modulation scheme. If we chose an alphabet of M constellation points on I and Q modulator, we can then encode $\log_2 M$ bits of information on each symbol therefore,

$$\text{Bits transmitted/ Symbol} = \log_2(M) \quad (15)$$

Such passband PAM (Pulse Amplitude Modulated) systems are fundamentally limited to symbol/sec/Hz, which makes the overall capacity $\log_2(M)$ bits/sec/Hz (bps/Hz). Adding an excess bandwidth factor α to allow for practical raised cosine pulse shaping gives the expression for overall channel capacity as:

$$R = \frac{\log_2(M)}{1+\alpha} \text{ bps/Hz} \quad (16)$$

3.3 Power efficiency

Power efficiency of a modulation scheme is determined by how information may be transmitted over a given channel with a given modulation scheme using a limited power. The channel capacity, however, is not without bound. Greater alphabet size (M) implies smaller spacing between constellation points for a given output power, reducing the tolerance of noise or distortion. This effect is expressed in the well-known Shannon-Hartley Limit shown in (19):

$$N_b = B \log_2 (1 + \text{SNR}) \quad (17)$$

where N_b = Number of bits/symbol, and

B = Bandwidth

$$\text{SNR} = \text{Signal to Noise Ratio} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$$

Therefore, for a particular power used in transmitting a signal, the rate i.e. channel capacity at which the signal is been transmitted can be known and for a given channel capacity the power required to transmit the signal can be known.

4. Discussion of the Results

The simulation was carried out on the MATLAB environment, in which four (4) PSK schemes were used i.e. BPSK, QPSK, 8-PSK and 16-PSK. MATLAB was used to simulate the average bit error rate (BER), bandwidth efficiency, symbol error rate (SER) and power efficiency for each of the schemes.

4.1 Bandwidth Efficiency

From Figure 2 it can be deduced that the higher the number of constellation point of a given modulation scheme, the more bandwidth efficient it would be it is and it is also shown that the BPSK scheme modulate only one bit at a time because it has only two constellation points. QPSK is twice more bandwidth efficient than BPSK because it modulate two bits at a time and 8-PSK scheme is three times bandwidth efficient than BPSK. For 16-PSK scheme, it is twice bandwidth efficient than QPSK and four times bandwidth efficient than BPSK because it modulates four bits at a time.

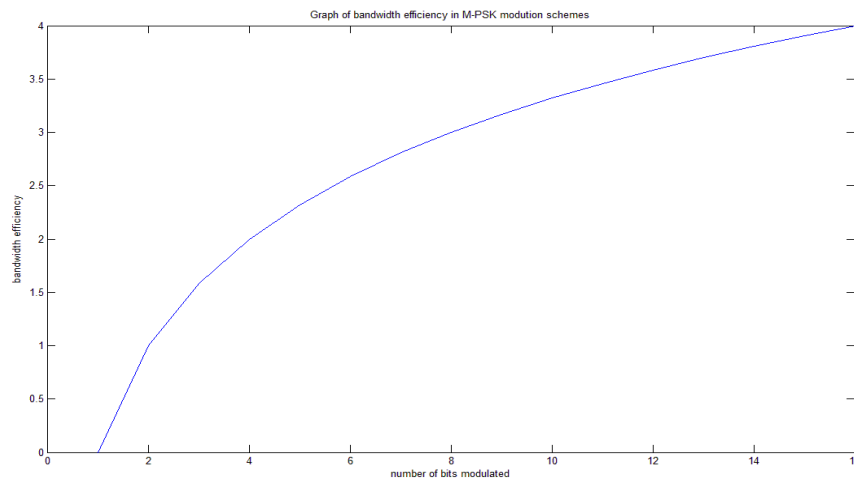


Figure 4.1: Bandwidth efficiency of M-PSK modulation schemes.

4.2 Bit Error Rate [BER]

Figure 3 shows the error performance for M-PSK modulation schemes. For all schemes, the BER decreases monotonically with increase in E_b/N_0 which is the carrier power. Therefore, it shows that increasing the carrier power of M-PSK modulation schemes will decrease the error performance of each schemes and it also shows that the error performance of BPSK and QPSK are approximately equal.

Furthermore, at higher M-PSK schemes, more carrier power is needed to modulate the signal in order to give low error performance; and in order to maintain the error performance of a scheme, the carrier power must be increased.

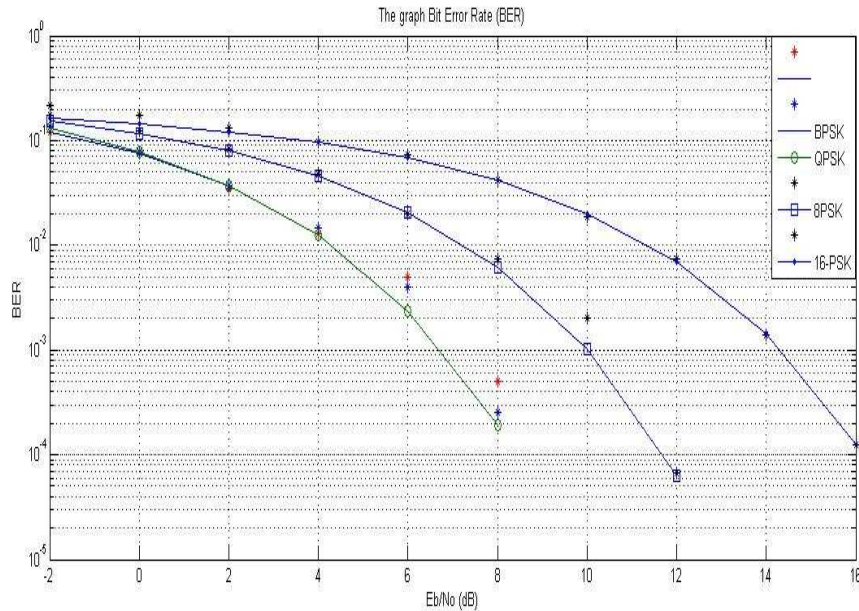


Figure 3: The Bit Error Rate of M-PSK modulation schemes.

4.3 Symbol Error Rate [SER]

This is related to BER but it is the error received at the sink per symbol; and it can also be minimised by increasing the carrier power. Figure 4 shows the trade-offs between the modulating carrier power and the SER but unlike BER the SER for BPSK and QPSK are distinct. At higher M-PSK schemes, high carrier power is required to minimise the error performance.

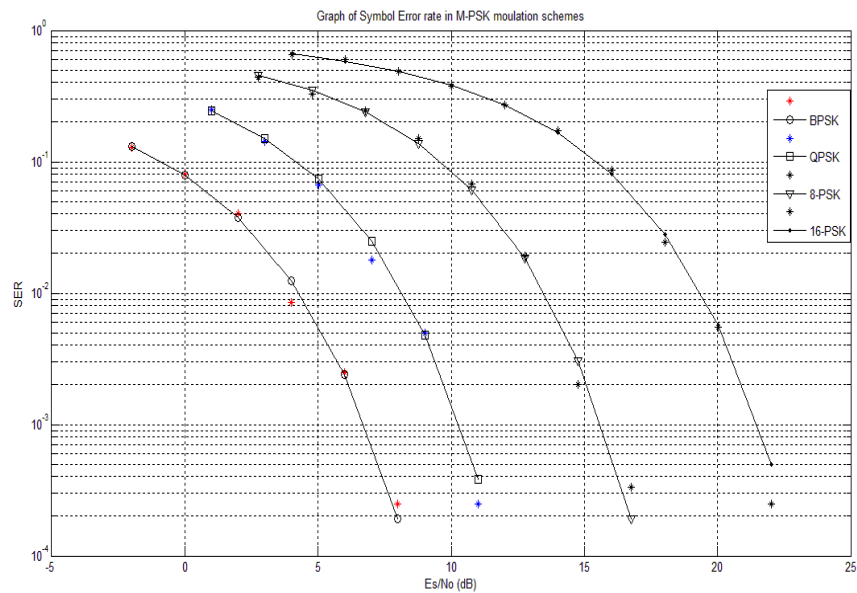


Figure 4: The Symbol Error Rate of M-PSK modulation schemes.

4.4 Power Efficiency

The power efficiency of M-PSK modulation schemes that was described previously is the trade-off that exists between carrier power and the error performance of each scheme. In Figure 5, the relationship between the carrier power and bandwidth efficiency was given for M-PSK modulation schemes.

From the figure, it can be deduced that modulation with low power make a given modulation scheme to be more bandwidth efficient i.e. the lower the carrier power, the higher or more bandwidth efficient it will be.

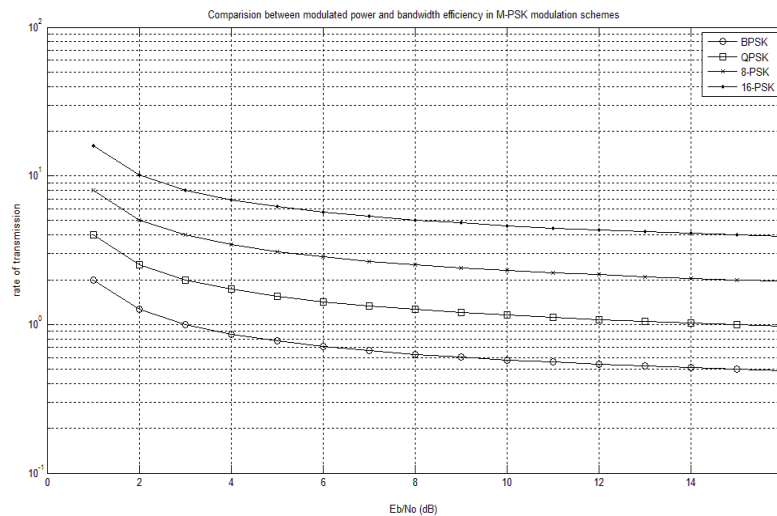


Figure 5: The relationship between the power efficiency and bandwidth efficiency.

5. Conclusion

Four (4) modulation methods were considered in this paper. These are the BPSK, QPSK, 8-PSK and 16-PSK. The average bit error rate (BER), bandwidth efficiency, symbol error rate (SER) and power efficiency were simulated for each of the methods. Findings show that lower end phase shift keying schemes can be used for purposes that involve low error performance and minimum power but low bandwidth efficient, while the higher schemes suit purposes that require higher bandwidth efficiency but are low power efficient.

Areas of further research include

- 1.) investigating other metrics other than the ones carried out in this paper, e.g bit energy-to-noise (E_b/N_0), noise immunity, channel capacity, etc.
- 2.) comparing the phase shift keying schemes with other modulation schemes like the frequency shift keying and amplitude shift keying

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Authors' biography

Adeleke, Oluseye A. obtained his Bachelor and Master's degrees in Electronic & Electrical Engineering from the Ladoke Akintola University of Technology, Ogbomoso, Nigeria. He is a member of the Nigerian Society of Engineers (NSE) and registered with the Council for the Regulation of Engineering in Nigeria (COREN).

He is currently working on his PhD in the area of Cooperation in wireless networks at the Universiti Sains Malaysia, Pulau Pinang, Malaysia. His areas of interests include wireless adhoc networks and application of game theory to wireless communication. Tel: +60142577508; e-mail: aoa11_eee129@student.usm.my

Abolade, Robert O. is presently with the Department of Electronic & Electrical Engineering, Ladoke Aintola University of Technology, Ogbomoso, Nigeria. His area of research interest is the Adaptive Multiple-In Multiple-Out (MIMO) Antennas. E-mail: robjoke2001@yahoo.com